Advanced Quantitative Research Methods

Bayesian Inference

Advanced module A12 — optional self-study

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Slides, lab notebooks and data:
github.com/ChrisW09/Advanced-Quantitative-Research-Methods

Where this module fits

Course chapter / module	What this module builds on it
Ch. 0 — statistics refresher	Confidence intervals and p-values — here they get rivals
Ch. 3 — linear regression	The likelihood; Bayesian regression reuses the whole fit
Ch. 4 — classification	Bayes' theorem itself (LDA already used it for labels)
Ch. 6 — regularisation	Ridge *is* a Bayesian estimate in disguise — proved here

The other way to be uncertain

The whole taught course is frequentist: parameters are fixed, data are random. This module flips it — **treat the parameter as the random thing**, update beliefs with data, and read off statements the frequentist machinery cannot make: "the probability that p exceeds 0.25 is 0.93."

One of twelve optional advanced modules; self-study. Every number on these slides is reproduced by `make_figures.py` in the module folder.

Contents

1. The Problem
2. The Beta-Binomial Engine
3. Gaussian Means & Shrinkage
4. Computation: MCMC
5. Regression & Decisions
6. Python Lab
7. Summary

Optional and advanced material is collected in the **appendix** at the end of the deck.

Notation in this module

Symbol	Meaning
θ	the unknown parameter, now carrying a distribution
$\pi(\theta)$	the prior: belief before the data
$L(\theta) = p(\text{data} \mid \theta)$	the likelihood (Chapters 3–4's object)
$\pi(\theta \mid \text{data})$	the posterior: belief after the data
Beta(a, b)	the workhorse prior for a probability
CrI	credible interval (posterior probability interval)
MAP	maximum a posteriori: the posterior's peak
τ	prior standard deviation (how sure you are beforehand)

Sources and references

Primary textbook

James, Witten, Hastie, Tibshirani & Taylor (2023), *An Introduction to Statistical Learning, with Applications in Python*, Springer. — Supplies the likelihoods, ridge, and the datasets (`Advertising.csv`).

Canonical further reading (optional)

Gelman et al., *Bayesian Data Analysis* (3rd ed., 2013); McElreath, *Statistical Rethinking* (2nd ed., 2020) — the standard references, one formal, one narrative.

Learning objectives

By the end of this module you will be able to:

- **State** Bayes' theorem for parameters and the roles of prior, likelihood and posterior.
- **Update** a Beta prior with binomial data by hand, and report a posterior mean and credible interval.
- **Explain** every posterior mean in this module as a *precision-weighted compromise* between prior and data.
- **Derive** ridge regression as the MAP estimate under a Gaussian prior — and translate $\lambda \leftrightarrow \tau$.
- **Implement** a Metropolis sampler from scratch and check it against the analytic answer.
- **Distinguish** credible from confidence intervals without garbling either.

Roadmap of this module

Believe, observe, update

1. The problem: what kind of probability a parameter can have.
2. The Beta–binomial engine: updating by hand, priors, credible intervals.
3. Gaussian means and shrinkage: precision weighting — and ridge unmasked.
4. Computation: when conjugacy runs out, walk (Metropolis).
5. Bayesian regression and decisions: `Advertising`, and an A/B call.
6. Python lab, summary, appendix.

Outline

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Two questions that sound the same

Frequentist (Ch. 0)	**Bayesian (this module)**
How would my estimate vary across imaginary repeated samples?	What should I believe about θ, given the one sample I have?
Parameter fixed, data random	Data fixed, parameter random
p-value, confidence interval	posterior, credible interval
"95% of such intervals cover"	"95% probability θ is in here"

Rule (Bayes' theorem for parameters)

$$\boxed{\pi(\theta \mid \text{data}) \;\propto\; L(\theta)\,\pi(\theta)} \qquad \text{posterior} \propto \text{likelihood} \times \text{prior}.$$

How to read this

Chapter 4's LDA already ran Bayes' theorem — on *class labels*. The only new move is to run it on *parameters*. The likelihood is the same object Chapters 3–4 maximised; the prior is new; the posterior is the answer.

Exercise A12.1 — Credible or confident? [Concept]

Exercise

A 95% *confidence* interval for a conversion rate is $[0.19, 0.31]$; a 95% *credible* interval from the same data is $[0.19, 0.31]$ as well.

1. Write the one-sentence claim each interval is entitled to make.
2. A manager says: "there is a 95% chance the true rate is between 0.19 and 0.31." Which interval licenses that sentence?
3. Why do the two intervals often nearly coincide anyway?

Solution — Exercise A12.1

Solution

Part 1. Confidence: "this interval was built by a recipe that captures the true rate in 95% of repeated samples" — a statement about the *procedure*. Credible: "given the data and the prior, the posterior probability that the rate lies in $[0.19, 0.31]$ is 0.95" — a statement about *this* interval.

Part 2. Only the credible interval. The manager's sentence assigns a probability to the parameter, which frequentist logic forbids: once computed, a confidence interval either contains the fixed truth or does not. (Everyone reads confidence intervals the manager's way — Chapter 0 flagged the misreading; here it becomes a licensed statement.)

Part 3. With a flat-ish prior and decent n, the posterior is essentially the normalised likelihood, so both intervals track the same curvature — they agree numerically while meaning different things. The divergence appears when priors are informative or n is small.

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Conjugacy: the update you can do in your head

Rule (Beta–binomial conjugacy)

$$\text{prior } \mathrm{Beta}(a, b) \;+\; y \text{ successes in } n \text{ trials} \;\Longrightarrow\; \boxed{\text{posterior } \mathrm{Beta}(a + y,\; b + n - y)}$$

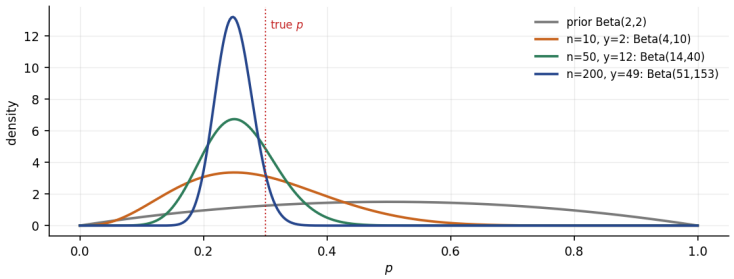
How to read this

The prior acts like $a+b$ *imaginary trials*, a of them successes. Updating is bookkeeping: add real successes to a, real failures to b. Posterior mean $= \frac{a+y}{a+b+n}$ — always *between* the prior mean and the sample proportion.

A conversion rate, true $p = 0.3$

Prior $\mathrm{Beta}(2, 2)$. After $n = 10$ ($y = 2$): $\mathrm{Beta}(4, 10)$, mean 0.286. After $n = 50$ ($y = 12$): $\mathrm{Beta}(14, 40)$, mean 0.259. After $n = 200$ ($y = 49$): $\mathrm{Beta}(51, 153)$, mean 0.250, 95% CrI [0.193, 0.312].

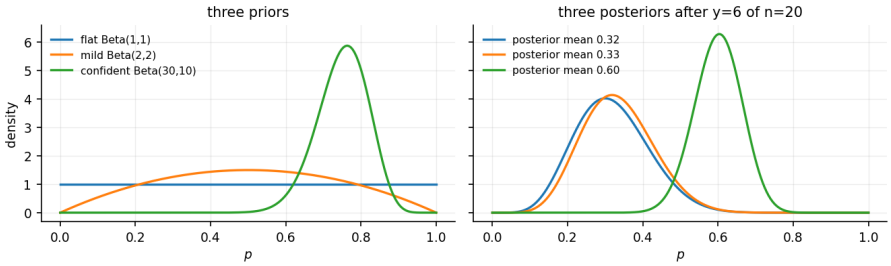
Watching belief sharpen



Data overwhelm the prior — at the data's own pace

The gentle Beta$(2, 2)$ hump narrows steadily; by $n = 200$ the posterior is tight around the sample rate and the prior's four imaginary trials are outvoted $50 : 1$. The truth (0.3) sits inside every credible interval along the way.

Three priors, one dataset



Priors matter exactly as much as they claim to

Same data ($y = 6$ of $n = 20$). Flat prior $\to$ mean 0.318; mild Beta$(2, 2)$ $\to$ 0.333; a confident-but-wrong Beta$(30, 10)$ (forty imaginary trials at 75%) $\to$ 0.600 — the data are outvoted 2:1. A prior is a claim with a stake; make it inspectable, and show the sensitivity.

Exercise A12.2 — One update by hand [Math]

Exercise

Prior Beta$(2, 2)$ on a click-through rate; you observe 7 clicks in 10 impressions.

1. Write down the posterior distribution.
2. Compute the prior mean, the MLE, and the posterior mean.
3. Show that the posterior mean is a weighted average $w \cdot \text{MLE} + (1 - w) \cdot \text{prior mean}$, and identify w.

Solution — Exercise A12.2

Solution

Part 1. $\text{Beta}(2+7,\ 2+3) = \text{Beta}(9, 5)$.

Part 2. Prior mean $= 2/4 = 0.5$; MLE $= 7/10 = 0.7$; posterior mean $= 9/14 \approx 0.643$.

Part 3. With $a + b = 4$ prior trials and $n = 10$ real ones,

$$\frac{a+y}{a+b+n} = \underbrace{\frac{n}{n+a+b}}_{w=10/14} \cdot \frac{y}{n} + \underbrace{\frac{a+b}{n+a+b}}_{1-w=4/14} \cdot \frac{a}{a+b} = \tfrac{10}{14}(0.7) + \tfrac{4}{14}(0.5) = 0.643.$$

The weight on the data is *sample size against imaginary sample size* — the entire prior-vs-data politics of this module in one fraction.

Common mistake

Reporting the MAP ($8/12 = 0.667$) as "the Bayesian estimate" without saying so. Mean, median and MAP all summarise the posterior; for skewed posteriors they differ, and the honest report is the whole interval.

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The Gaussian mean: precision-weighted compromise

Rule (known σ, prior $\mu \sim N(\mu_0, \tau^2)$)

$$\text{posterior mean} = w\,\bar{x} + (1-w)\,\mu_0, \qquad w = \frac{n/\sigma^2}{n/\sigma^2 + 1/\tau^2}$$

$$\text{posterior variance} = \left(n/\sigma^2 + 1/\tau^2\right)^{-1}.$$

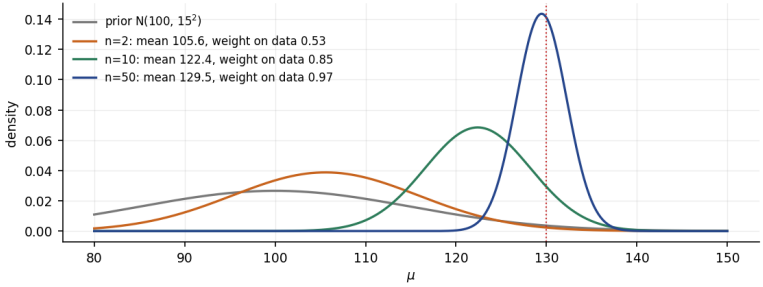
How to read this

Precisions (inverse variances) *add*; means combine *weighted by precision*. The data's vote grows linearly in n; the prior's vote is fixed — the Beta–binomial story again, in Gaussian dress.

Prior $N(100, 15^2)$, $\sigma = 20$, truth 130

$n = 2$: mean 105.6, weight on data 0.53. $n = 10$: 122.4, weight 0.85. $n = 50$: 129.5, weight 0.97 — the prior fades on schedule.

The prior fading, drawn



Shrinkage is not a bug

At $n = 2$ the posterior barely moves off the prior — and that is the *correct* behaviour: two noisy observations should not overturn a considered belief. Every shrinkage estimator in Chapter 6 is this picture wearing an optimisation costume, as the next slide proves.

Ridge regression, unmasked

Claim

Ridge (Ch. 6) is the **MAP estimate** of a linear model under the prior $\beta_j \sim N(0, \tau^2)$, with $\boxed{\lambda = \sigma^2/\tau^2}$.

The two-line proof

Maximising the log-posterior = log-likelihood + log-prior:

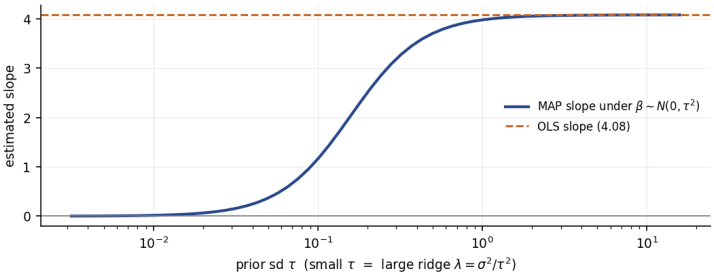
$$-\tfrac{1}{2\sigma^2}\textstyle\sum_i (y_i - x_i^\top \beta)^2 \;-\; \tfrac{1}{2\tau^2}\textstyle\sum_j \beta_j^2 \;\;\propto\;\; -\Big[\textstyle\sum_i (y_i - x_i^\top \beta)^2 + \tfrac{\sigma^2}{\tau^2}\textstyle\sum_j \beta_j^2\Big],$$

which is exactly the ridge criterion with $\lambda = \sigma^2/\tau^2$.

Chapter 6's tuning dial, translated

A large penalty λ *is* a tight prior (τ small). Cross-validating λ empirically chooses how strong a prior the data can support. (The lasso? A Laplace prior — same construction.)

The dictionary, drawn on Advertising



Read the dial

Standardised `sales`-on-`TV` slope: OLS says 4.08. As the prior tightens ($\tau \downarrow$), the MAP slope shrinks toward 0 — at $\tau = 0.1$ (i.e. $\lambda = 500$) it is 1.16. Chapter 6's whole coefficient-path figure is this curve, one predictor at a time.

Exercise A12.3 — Precision weighting by hand [Math]

Exercise

Quality control believes a machine fills bottles at $\mu_0 = 100$ml ($\tau = 15$). Measurement noise is $\sigma = 20$. A morning sample gives $n = 4$, $\bar{x} = 120$.

1. Compute the posterior mean and standard deviation of μ.
2. The engineer says "the sample says 120, so recalibrate to 120." What does the posterior say instead, and why?
3. How large would n (at $\bar{x} = 120$) have to be for the weight on the data to reach 0.9?

Solution — Exercise A12.3

Solution

Part 1. Precisions: prior $1/\tau^2 = 1/225 \approx 0.00444$; data $n/\sigma^2 = 4/400 = 0.01$. Weight on data $w = 0.01/0.01444 \approx 0.692$. Posterior mean $= 0.692 \times 120 + 0.308 \times 100 \approx 113.8$; posterior sd $= \sqrt{1/0.01444} \approx 8.3$.

Part 2. Split the difference toward 113.8: with only four noisy measurements (SE $= 20/2 = 10$), a 20ml jump from a well-calibrated prior is as likely to be sampling noise as drift. The posterior *is* the rational compromise — and its sd (8.3) says a follow-up sample is worth taking before touching the machine.

Part 3. Need $\frac{n/400}{n/400+1/225} = 0.9 \Rightarrow n/400 = 9/225 \Rightarrow n = 16$.

Common mistake

Adding standard deviations instead of precisions. Only *inverse variances* add; the posterior sd is always *smaller* than both the prior's and the data's alone.

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When conjugacy runs out, walk

Real models rarely have closed-form posteriors. The general tool: build a random walk whose long-run visiting frequencies *are* the posterior.

Algorithm (Metropolis)

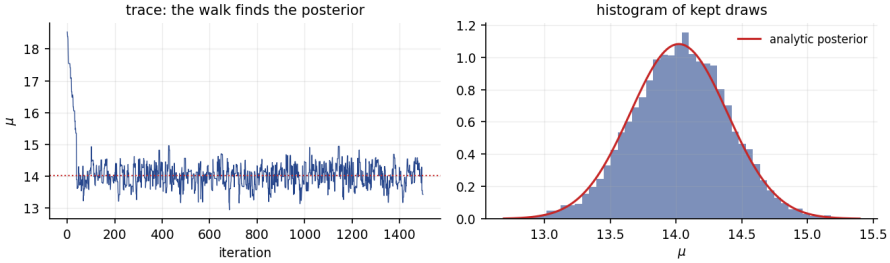
From current θ: (1) propose $\theta' = \theta + N(0, s^2)$; (2) accept with probability $\min\big(1,\ \pi(\theta' \mid \text{data})/\pi(\theta \mid \text{data})\big)$, else stay; (3) record, repeat. Discard the warm-up ("burn-in"), keep the rest.

Why it works, in one sentence

Moves uphill are always taken and moves downhill sometimes — calibrated so the walk spends time in each region *in proportion to its posterior probability*; only posterior *ratios* are needed, so the intractable normalising constant cancels.

Like A2's Shapley values and A3's conformal intervals, we build it in 15 lines; the production tools are `PyMC`, `Stan`, `NumPyro`.

Metropolis on real data: the mean of `Advertising` sales



Sampler vs analytic truth

20,000 steps, acceptance rate 0.62, burn-in 2,000. Posterior mean 14.04 vs analytic 14.02; posterior sd 0.367 vs analytic 0.369. The trace (left) starts at a deliberately wrong value and finds the posterior within a few hundred steps.

Exercise A12.4 — Reading a sampler [Concept]

Exercise

Three colleagues run Metropolis on the same posterior and show you their diagnostics:

1. Colleague A: acceptance rate 0.99; the trace drifts in slow waves.
2. Colleague B: acceptance rate 0.02; the trace is long flat plateaus.
3. Colleague C: acceptance rate 0.55; the trace is a fuzzy caterpillar.

Diagnose each, name the shared root cause for A and B, and state the fix.

Solution — Exercise A12.4

Solution

A — steps too small. Nearly every proposal is accepted because it barely moves; the chain explores the posterior like a snail, and consecutive draws are almost copies. Estimates converge, but painfully slowly.

B — steps too large. Proposals leap into low-probability territory and are almost always rejected; the chain freezes for hundreds of iterations. Same disease as A — *high autocorrelation in the chain* — reached from the opposite direction.

C — healthy. Mid-range acceptance with fast mixing; rules of thumb put the sweet spot around 0.2–0.5 for random-walk Metropolis (our run: 0.62, acceptably brisk for one dimension).

Fix. Tune the proposal scale s toward the sweet spot; then confirm with the trace and by running a second chain from a different start and checking both agree.

Common mistake

Judging a sampler by acceptance rate alone. It is a symptom, not the disease: the disease is autocorrelation, and the trace plot (plus multiple chains) is the diagnostic that sees it.

Extended Exercise A12.1 — Your own sampler [Python]

Extended exercise (15 min)

Implement Metropolis from scratch for the mean of `Advertising` sales (treat σ as the sample sd; prior $\mu \sim N(0, 100^2)$):

1. Write the log-posterior and the accept/reject loop; run 20,000 steps at proposal scale 0.5 from the (bad) start $\bar{x} + 5$.
2. Report acceptance rate, posterior mean and sd after burn-in; compare with the analytic values (14.02, 0.369).
3. Re-run with proposal scales 0.01 and 50. Connect what you see to Exercise A12.4.

Solution — Extended Exercise A12.1 (1/2) — code

```
import numpy as np, pandas as pd
x = pd.read_csv("Advertising.csv", index_col=0)["sales"].to_numpy()
rng = np.random.default_rng(2024)
sigma, tau, B, s = x.std(ddof=1), 100.0, 20_000, 0.5

def logpost(mu): # log posterior, up to a constant
    return -0.5*((x - mu)**2).sum()/sigma**2 - 0.5*mu**2/tau**2

chain, cur = np.empty(B), x.mean() + 5
lp = logpost(cur); acc = 0
for i in range(B):
    prop = cur + rng.normal(0, s)
    lp_prop = logpost(prop)
    if np.log(rng.uniform()) < lp_prop - lp: # uphill: always; downhill: sometimes
        cur, lp, acc = prop, lp_prop, acc + 1
    chain[i] = cur
kept = chain[2000:]
print(f"accept {acc/B:.2f}, mean {kept.mean():.2f}, sd {kept.std(ddof=1):.3f}")
```

Solution — Extended Exercise A12.1 (2/2)

Solution — parts 2 and 3

Part 2. Acceptance ≈ 0.62; posterior mean 14.04, sd 0.367 — against analytic 14.02 and 0.369. A dozen lines of NumPy reproduce the exact answer to two decimals; that agreement *is* the test that the sampler is correct.

Part 3. At $s = 0.01$ the acceptance rate rockets toward 1 and the trace crawls (colleague A); at $s = 50$ acceptance collapses toward 0 and the trace is plateaus (colleague B). Both chains still *eventually* target the right posterior — the theory guarantees the destination, tuning buys the arrival time.

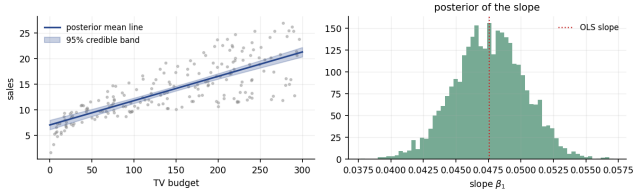
Common mistake

Using $\pi(\theta')/\pi(\theta)$ on the raw (non-log) scale. With $n = 200$ the likelihood underflows `float64` long before the ratio is formed — always accept/reject on the *log* scale, as the code does.

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Bayesian simple regression: `sales` on `TV`



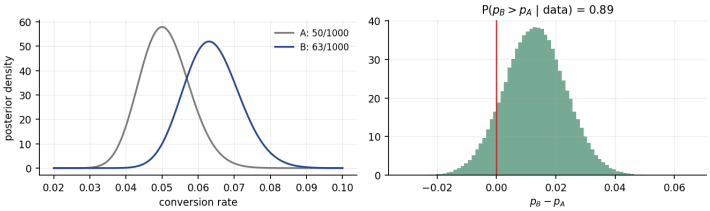
The posterior of a Chapter 3 slope

Weak $N(0, 10^2)$ priors on both coefficients. Posterior slope: mean 0.0476, sd 0.0027 — numerically the OLS fit (0.0475) with its standard error, *reinterpreted*: the left panel's band is a 95% credible band of regression lines drawn from the posterior.

Weak priors reproduce Chapter 3 — by design

With flat-ish priors, Bayesian regression rediscovers least squares; the machinery pays when priors carry information — small n, correlated predictors (ridge!), hierarchies of related groups.

A decision, not a *p*-value: the Bayesian A/B test



A: 50/1000, **B:** 63/1000

Flat priors, two independent Beta posteriors, 200,000 paired draws: $\Pr(p_B > p_A \mid \text{data}) = 0.895$ — a directly decision-shaped number. (The two-sided frequentist test at these counts does not clear 0.05; both summaries are honest, they answer different questions.)

Common mistake

Believing posteriors are immune to module A1's peeking problem: if you *stop when the posterior looks good*, the long-run error rate of that *policy* still inflates — design the stopping rule, whatever your school.

Extended Exercise A12.2 — The Bayesian A/B memo [Integrative]

Extended exercise (15 min)

Marketing ran the experiment above (A: 50/1000, B: 63/1000) and asks for a recommendation memo.

1. Reproduce $\Pr(p_B > p_A \mid \text{data})$ by simulation with flat priors. State the number with its meaning in one sentence.
2. Compute the posterior of the *lift* $p_B - p_A$ and report a 95% credible interval. Does it contain 0, and does that contradict part 1?
3. Switching costs EUR 2,000; each conversion is worth EUR 5 on 100,000 annual visitors. Turn the posterior into an expected-value decision.
4. State one sentence of prior sensitivity: what would a sceptical prior (e.g. Beta$(50, 950)$ on both arms) do to the conclusion?

Solution — Extended Exercise A12.2

Solution

Part 1. Draw from Beta$(51, 951)$ and Beta$(64, 938)$, compare: $\Pr(p_B > p_A) \approx 0.895$ — "given the data, B is better with probability about 0.9."

Part 2. The lift's posterior is centred near 0.013 with 95% CrI roughly $[-0.007, 0.033]$ — it *does* straddle 0. No contradiction: the CrI asks "what values are plausible", part 1 asks "which arm is more likely better"; a 0.895 direction is compatible with an interval touching zero.

Part 3. Expected lift $\approx 0.013 \times 100{,}000 \times$ EUR 5 = EUR 6,500/year against EUR 2,000 once; even weighting the downside draws, expected value favours switching — recommend B, note the $\sim 10\%$ chance it is not better, and (module A1) confirm with a pre-registered follow-up if the cost were larger.

Part 4. A sceptical Beta$(50, 950)$ prior (five percent, held with 1000 trials' conviction) roughly halves the estimated lift and pulls $\Pr(p_B > p_A)$ toward 0.8 — the direction survives, the enthusiasm moderates. Report both runs.

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Python lab — run it live

Reproduce every number: `make_figures.py`

Every number on these slides is computed by `make_figures.py` in this module's folder, from the bundled `Advertising.csv` and seeded simulations — run it and read it alongside the deck:

- the Beta–binomial updates ($\text{Beta}(2,2) \to \text{Beta}(51,153)$) and the three-priors picture;
- the precision-weighted Gaussian posteriors ($w = 0.53 \to 0.97$);
- the from-scratch Metropolis run (14.04 vs analytic 14.02);
- Bayesian regression on `Advertising` and the ridge–MAP dial;
- the A/B posterior ($\Pr(p_B > p_A) = 0.895$).

A companion notebook in the course lab format is in preparation; until it lands, the script *is* the lab — it runs top to bottom in the course environment with no extra packages.

The whole module in twelve lines

```python
import numpy as np
from scipy import stats
rng = np.random.default_rng(2024)

# belief -> data -> updated belief, all in one conjugate line
a, b = 2, 2 # prior Beta(2,2): four imaginary trials
y, n = 49, 200 # the data
post = stats.beta(a + y, b + n - y)
print(f"posterior mean {post.mean():.3f}, "
      f"95% CrI [{post.ppf(0.025):.3f}, {post.ppf(0.975):.3f}]")

# and the sentence the frequentist interval cannot say:
print(f"P(p > 0.25 | data) = {1 - post.cdf(0.25):.2f}")
```

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Module A12 in one slide

What we accomplished

A second, coherent grammar for uncertainty — run on the course's own likelihoods and datasets.

- Posterior $\propto$ likelihood $\times$ prior — Chapter 4's theorem, aimed at parameters.
- Beta–binomial updating by hand: Beta$(2, 2)$ plus 200 trials $\rightarrow$ Beta$(51, 153)$, CrI $[0.193, 0.312]$.
- Every posterior mean is a precision-weighted compromise; the data's weight grows as $n/(n + \text{imaginary } n)$.
- Ridge unmasked: MAP under $N(0, \tau^2)$ priors, $\lambda = \sigma^2/\tau^2$ — Chapter 6 was Bayesian all along.
- Metropolis from scratch: 14.04 ± 0.367 against analytic 14.02 ± 0.369.
- Decision-shaped output: $\Pr(p_B > p_A \mid \text{data}) = 0.895$ — with the peeking caveat imported from A1.

Key formulas at a glance

Bayes for parameters $\pi(\theta \mid \text{data}) \propto L(\theta)\,\pi(\theta)$.

Beta–binomial $\text{Beta}(a,b) + (y,n) \to \text{Beta}(a{+}y,\ b{+}n{-}y)$; mean $(a{+}y)/(a{+}b{+}n)$.

Shrinkage weight posterior mean $= w\,\hat{\theta}_{\text{data}} + (1{-}w)\,\theta_{\text{prior}}$, $w = n/(n+a+b)$ (Beta) or $\frac{n/\sigma^2}{n/\sigma^2+1/\tau^2}$ (Gaussian).

Gaussian posterior precision $= n/\sigma^2 + 1/\tau^2$; precisions add, never sds.

Ridge as MAP $\hat{\beta}_{\text{ridge}} = \arg\max$ posterior under $\beta_j \sim N(0,\tau^2)$, with $\lambda = \sigma^2/\tau^2$.

Metropolis accept θ' w.p. $\min\{1, \pi(\theta' \mid \text{data})/\pi(\theta \mid \text{data})\}$ — ratios only, log scale in code.

Vocabulary checklist

Term	Meaning
prior	belief about θ before the data, as a distribution
likelihood	$p(\text{data} \mid \theta)$ — Chapters 3–4's fit criterion
posterior	updated belief; the module's universal output
conjugate prior	family the posterior stays inside (Beta for binomial)
credible interval	interval holding stated posterior probability
MAP	posterior mode; ridge and lasso are MAPs
shrinkage	posterior pulled from data toward prior
burn-in	discarded warm-up of an MCMC chain
acceptance rate	share of Metropolis proposals taken; tune toward mid-range
prior sensitivity	re-running with other priors; always reported

Decision rules of thumb

- Small n, real prior knowledge, or a decision to make $\Rightarrow$ Bayesian machinery earns its keep. Large n, flat priors $\Rightarrow$ it will agree with Chapter 0; use whichever is cheaper.
- Choose priors you can defend out loud, in units of imaginary observations — and always show a sensitivity run.
- Report the whole posterior (or its CrI), not just a point; say *which* point if you must give one.
- Conjugate first, grid second, MCMC when you must; always check a sampler against a case with a known answer.
- Trace plot + second chain before trusting any MCMC number.
- Probability statements about parameters: credible intervals only — never launder a confidence interval into one.

Common pitfalls

Don't

1. Read a confidence interval as "95% probability the parameter is inside" — that sentence needs a posterior.
2. Hide an informative prior (Beta$(30, 10)$ moved the answer from 0.32 to 0.60 on the same data).
3. Add standard deviations where only precisions add.
4. Compute likelihood ratios off the log scale ($n = 200$ already underflows).
5. Trust a chain by its acceptance rate alone — plateaus and snails both lie.
6. Peek-and-stop on a posterior and claim the error rate of a fixed-n design.

Self-check questions

Quick self-assessment

1. Prior Beta$(2,2)$, then 7 successes in 10 trials: posterior, posterior mean, and the weight on the data?
2. Prior $N(100, 15^2)$, $\sigma = 20$, $n = 4$, $\bar{x} = 120$: posterior mean?
3. State the ridge–Bayes dictionary in one equation.
4. Your Metropolis run has acceptance 0.99 and a slowly drifting trace. Diagnosis and fix?
5. $\Pr(p_B > p_A \mid \text{data}) = 0.895$ while the 95% CrI for $p_B - p_A$ includes 0. Contradiction?
6. What is the one sentence a credible interval may say that a confidence interval may not?

Answers

(1) Beta$(9,5)$; $9/14 \approx 0.643$; $w = 10/14 \approx 0.71$. (2) $w \approx 0.69$, mean ≈ 113.8. (3) $\lambda = \sigma^2/\tau^2$ under $\beta_j \sim N(0, \tau^2)$. (4) Proposal scale too small — enlarge s toward acceptance $\sim$ 0.2–0.5, confirm on the trace. (5) No: one is a statement about direction, the other about plausible magnitudes; both are read off the same posterior. (6) "Given the data, the parameter lies in this interval with probability 0.95."

Five things to remember

Module A12 in five lines

1. **Posterior $\propto$ likelihood $\times$ prior** — one line, the whole framework.
2. **Every estimate is a compromise weighted by precision** — the data's weight is n against the prior's imaginary n.
3. **Ridge is a prior in disguise:** $\lambda = \sigma^2/\tau^2$ — Chapter 6 has been Bayesian all along.
4. **MCMC trades formulas for a walk** — and a from-scratch Metropolis matched the analytic answer to two decimals.
5. **Priors are claims with stakes:** state them, price them in imaginary trials, and show the sensitivity run.

What's next

- **Module A1 — Randomised Controlled Trials**: the A/B decision slide is A1's experiment wearing posterior clothes — and A1's peeking warning survives the change of school.
- **Chapter 6, re-read**: the ridge and lasso paths are prior strength dialled from ∞ to 0; the Bayesian reading explains *why* shrinking helps when p is large.
- **Module A3 — Conformal Prediction**: the frequentist route to calibrated uncertainty — guarantees without priors, at the price of answering only about *predictions*. Reading A3 and A12 together maps the whole uncertainty landscape.

Appendix — what is in here, and why

Nothing in the appendix is needed to follow the module: each slide is a formal derivation, a longer worked exercise, or a side topic. Read it when you want the full story, or when a later chapter sends you back here.

Contents of the appendix

- **The Beta–binomial update, derived** — two lines of algebra; moved out because the recipe, not the proof, carries the deck.
- **A conjugate-pairs reference table** — look-up material.
- **The posterior predictive** — forecasting a new observation, not a parameter; the bridge to Chapters 3 and A3.

The Beta–binomial update, derived

Prior density $\pi(p) \propto p^{a-1}(1-p)^{b-1}$; binomial likelihood $L(p) \propto p^{y}(1-p)^{n-y}$. Multiply:

$$\pi(p \mid y) \;\propto\; p^{y}(1-p)^{n-y} \cdot p^{a-1}(1-p)^{b-1} \;=\; p^{(a+y)-1}\,(1-p)^{(b+n-y)-1},$$

which is the kernel of $\mathrm{Beta}(a+y,\; b+n-y)$ — no integral was ever computed; recognising the *shape* suffices.

Why conjugacy is a big deal

The prior and posterior live in the same family, so updating is arithmetic on two counters (a, b) — streaming data update beliefs in $O(1)$, and the "imaginary trials" reading of the prior is literal.

Conjugate pairs at a glance

Likelihood	Conjugate prior	Posterior update
Binomial (n, p)	Beta(a, b)	$a+y,\ b+n-y$
Poisson (λ)	Gamma(a, b)	$a + \sum y_i,\ b + n$
Gaussian mean (known σ)	$N(\mu_0, \tau^2)$	precision-weighted (main deck)
Gaussian variance (known μ)	Inverse-Gamma	shape $+ n/2$
Multinomial	Dirichlet	add the category counts

How to use this table

Each row is the same story: the prior is phrased as pseudo-data, and updating adds real data to it. Module A4's exponential-family machinery is the deep reason this table exists: every exponential-family likelihood has a conjugate family of exactly this form.

The posterior predictive: betting on the next observation

For a new observation $\tilde{y}$, integrate predictions over the posterior:

$$p(\tilde{y} \mid \text{data}) = \int p(\tilde{y} \mid \theta)\, \pi(\theta \mid \text{data})\, d\theta.$$

The next conversion

After Beta$(51, 153)$: $\Pr(\tilde{y} = 1 \mid \text{data}) = E[p \mid \text{data}] = 51/204 = 0.25$ — but for the next *twenty* visitors the predictive is Beta-binomial, *wider* than Binomial$(20, 0.25)$, because parameter uncertainty rides along.

The bridge to prediction intervals

Chapter 3's prediction intervals and module A3's conformal sets solve this same problem in frequentist dress: honest forecasts must carry *both* noise and parameter uncertainty. The posterior predictive is the Bayesian statement of that requirement.